



AI Application & Society - UNIV*6090: Fall 2025

Module-2

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1.Introduction

It is undeniable that Computer Numerical Control (CNC) machines have had a large impact on the manufacturing industry. Starting in the 1950s, CNC machines helped automate many of the expensive, high precision machining operations [1]. Some of the products these machines help create include mobile phones, automotive parts, computer, and much more [1]. One of the complex functions of a CNC machine is the milling process, since the product being cut requires a specific size and shape with little room for error. Another issue is that the tools used for the milling process, which are also expensive, are worn down more after every machining run. Since these worn tools damage the material being cut during machining processes, they have to be replaced before they damage the material, but not before they are fully used.

One of the solutions to this optimization problem is machine learning models. Machine learning models have been widely used in the process of tool wear prediction [2]. There are many reasons for using ML models for tool wear prediction. Some benefits are that tool wear predictions can help maintain productivity, minimize costs, and minimize material waste [2]. Yet, tool wear prediction models tend to be sensitive to small variation and not easily generalizable [2,3]. The data used for the machine learning model training focus on a specific use case and tend to come from a single milling process [2]. As a result, it is an important task to analyze each specific use case when creating a tool wear prediction model. This paper contributes to the research landscape in the following ways:

- We show that feature ranking for tool wear prediction can reduce a set of 163 features to the 20 most important features
- We utilized 4 machine learning models and an ensemble model to predict tool wear. This ultimately resulted in the random forest machine learning model having the best performance with a 93.94% accuracy.
- We fine-tuned the amount of sensors used to result in a tool wear prediction model that only uses 1 sensor to optimize the cost.

- We split the wear of a tool into a finite number of classes and compared the optimal number of classes. The results indicated that 3 classes resulted in the highest accuracy of predictions while 8 classes had the finest granularity.

The rest of the paper will be organized as follows. In section 2 we discuss the methodology of the paper. This section will include a data related methodology such as a dataset description, data preprocessing, and feature ranking. This section will also include a methodology for the machine learning models such as the predictive models that we use and the evaluation metrics. In section 3 we analyze the results by comparing the models across differing amounts of sensors and differing amounts of classes for the target variable. In section 4 we conclude the paper with an overview of what was accomplished.

2. Methodology

This project focuses on predicting tool wear in milling operations using machine learning techniques applied to the NASA Milling Dataset. The study involves preprocessing multi-sensor data, ranking the most influential features, and developing multiple classification models including Random Forest, SVM, Logistic Regression, Gaussian Naïve Bayes, and an Ensemble model. Each model's performance was evaluated using accuracy, precision, recall, F1-score, and effectiveness of metrics. The goal was to identify the optimal model and sensor combination that delivers accurate tool wear prediction while minimizing complexity and cost.

2.1 Dataset Description

The NASA milling dataset was collected from a set of runs on a Matsuura machining center MC-510V [4]. The sensors used to collect the data were either an acoustic emission sensor (AE), a current sensor (smc), or a vibration sensor (vib). While there were three sensors, each of the sensors were measuring in two different configurations. For both the AE and vib sensors, they were collecting data from the table and the spindle, but for the smc sensor, there was an AC smc and DC smc. There were two materials being cut, either cast iron or stainless steel J45. The cutting speed was set to 826 rev/min. There were two different depths of cut (DOC), 1.5 mm and 0.75 mm. There were also two different feed rates, 0.25 mm/rev and 0.5 mm/rev. The workpiece was

483mm x 178mm x 51mm in size. The insert used was of type KC710 and there were 8 different cases done based on a combination of material, DOC, and feed rate. This experiment was repeated a second time to add up to the total of 16 cases with a varying number of runs. Table 1 below summarizes the configurations for each case.

Table 1: *Cutting parameters for the dataset provided in [4]*

Case Number	Depth of Cut (mm)	Feed (mm/rev)	Material
1	1.5	0.5	Cast iron
2	0.75	0.5	Cast iron
3	0.75	0.25	Cast iron
4	1.5	0.25	Cast iron
5	1.5	0.5	Steel
6	1.5	0.25	Steel
7	0.75	0.25	Steel
8	0.75	0.5	Steel
9	1.5	0.5	Cast iron
10	1.5	0.25	Cast iron
11	0.75	0.25	Cast iron
12	0.75	0.5	Cast iron
13	0.75	0.25	Steel
14	0.75	0.5	Steel
15	1.5	0.25	Steel
16	1.5	0.5	Steel

2.2 Data Preprocessing

Data preprocessing was an important step in this project to ensure data quality, consistency, and readiness for feature extraction and model training. The NASA Milling Dataset contains multi-sensor time series signals collected during titanium milling operations under varying cutting conditions. The raw data consisted of readings from vibration (vib), acoustic emission (AE), and spindle motor current (smc) sensors, each associated with tool wear progression measured as flank wear (VB).

Step 1: Data Merging and Synchronization

Sensor data were first merged into a unified dataset using time alignment and run identifiers. Each experimental run was treated as one observation, with aggregated features representing the

corresponding cutting operation. The VB value from each run was recorded as the target variable. Data from multiple sensor locations (e.g., spindle, table, and control modules) were combined to capture diverse tool–workpiece interaction dynamics.

Step 2: Feature Extraction.

From each sensor signal, a comprehensive set of domains (e.g., mean, RMS, variance, standard deviation, peak-to-peak) and frequency-domain features (e.g., spectral centroid, band power, roll-off, and flatness) was computed. This produced a total of 168 engineered features, each characterizing a specific aspect of signal energy, periodicity, or frequency content relevant to tool wear.

Step 3: Data Cleaning.

Missing or corrupted entries were identified and handled using conditional imputation or removal, depending on the proportion of missingness. Outliers were inspected through statistical thresholds and corrected or excluded to prevent model bias. Additionally, feature redundancy was checked using correlation analysis to ensure that multicollinearity did not distort feature importance estimates.

Step 4: Normalization.

To harmonize the feature scales and prevent dominance by variables with large numeric ranges, all features were normalized using z-score standardization as [5]:

$$z = \frac{x - \mu}{\sigma}$$

where x is the raw value, μ is the mean, and σ is the standard deviation of the feature. This ensured comparability across sensors and improved convergence for models sensitive to feature magnitude, such as SVM and Logistic Regression.

Step 5: Data Labeling.

The continuous VB values were later categorized into three discrete wear classes—Low Wear ($VB \leq 0.15$), Medium Wear ($0.15 < VB \leq 0.30$), and High Wear ($VB > 0.30$)—to facilitate classification-based predictive modeling. This stratification also helped balance the class distribution during training and evaluation.

The fully preprocessed dataset, consisting of 165 samples and 168 normalized features, was then used for feature ranking and predictive modeling as described in the subsequent sections.

2.3 Feature Ranking

Feature ranking was conducted to identify the most significant predictors of tool wear and to reduce model complexity by eliminating less informative variables. The combined dataset included both time domain and frequency domain features extracted from multiple sensors: vib, acoustic emission, and spindle current. In total, 168 features were analyzed across 165 samples, using the Random Forest Regressor model to quantify feature importance based on their contribution to predicting continuous VB values.

The feature selection process followed four key steps. Step 1 involved merging all sensor-derived datasets into a unified structure containing synchronized readings and corresponding VB measurements. Step 2 applied z-score normalization to standardize all feature values, preventing bias due to differing numerical scales across sensor modalities. Step 3 converted continuous VB measurements into three discrete wear classes—Low Wear ($VB \leq 0.15$), Medium Wear ($0.15 < VB \leq 0.30$), and High Wear ($VB > 0.30$)—to support stratified sampling and facilitate classification analysis. Finally, step 4 focused on ranking the features using the Random Forest algorithm, which provides an intrinsic measure of variable importance by averaging the decrease in impurity across decision trees.

The Random Forest model was configured with an 80-20 train-test split, stratified by VB class, and included 200 decision trees to ensure stable estimates of feature importance. An elbow method was used to determine an appropriate cutoff for selecting the most impactful features. The threshold value of 0.0134 corresponded to approximately 80% cumulative importance, beyond which additional features contributed minimally to prediction accuracy.

The feature ranking curve, shown in Figure 1 shows the distribution of importance values (blue points) along with the applied threshold (red dashed line) and selected feature count (green vertical line). A total of 20 top ranked features were selected for subsequent model training.

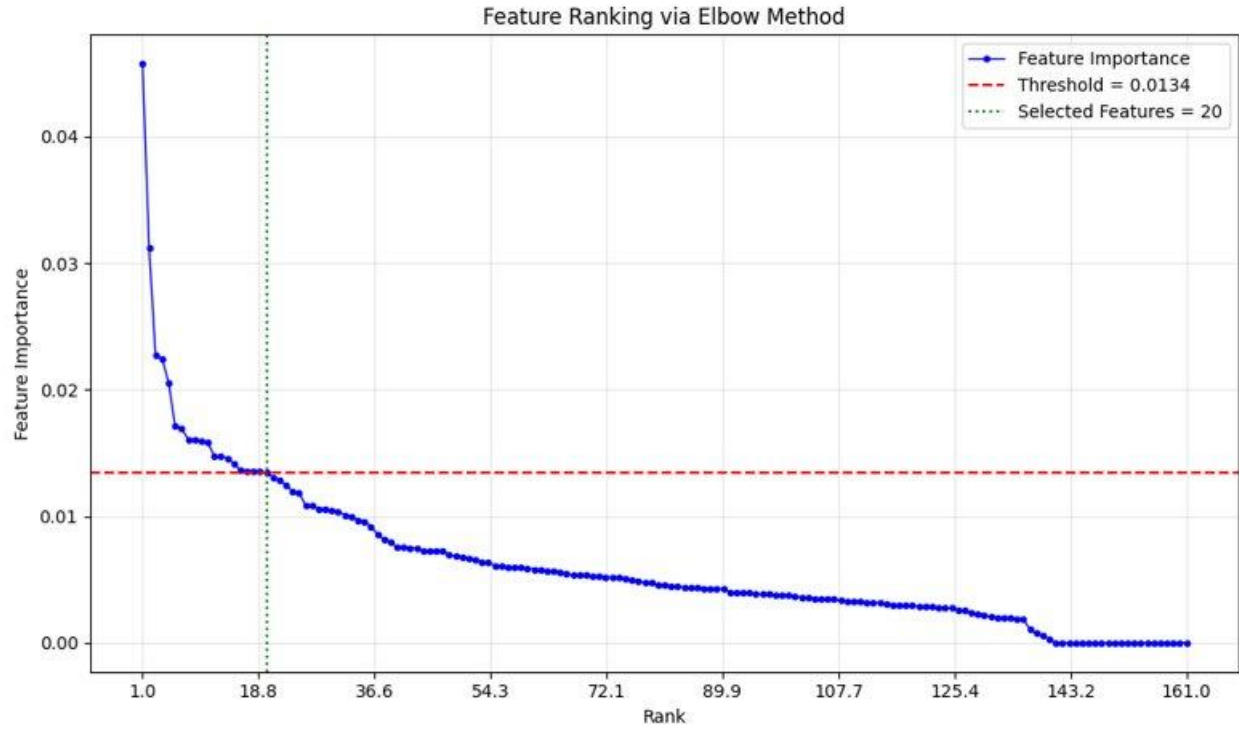


Figure 1: Feature ranking via Elbow Method showing threshold (0.0134) and top 20 selected features.

These top 20 features were predominantly derived from the vibration spindle (vib_spindle) and acoustic emission (AE_table) sensors, emphasizing their relevance in detecting tool wear progression. Representative examples include vib_spindle_rms, vib_spindle_mean, smcAC_p2p, and AE_table_mag_centroid. The relative importance of each sensor was then aggregated to evaluate the overall sensor-level contribution to wear prediction. The sensor important ranking is summarized in Table 2 below.

Table 2: Sensors and their aggregated importance (Bold font indicate highest importance score)

Sensor	Aggregated Importance
vib_spindle	0.3551
smcAC	0.1368
AE_table	0.1342
vib_table	0.1164
AE_spindle	0.1148

From these results, vib_spindle is the most dominant sensor as it had the highest aggregated importance value, contributing more than one third of the total importance score. Following vib_spindle, smcAC, AE_table and vib_table come afterwards. These four sensors were thus selected for subsequent model evaluation and combination testing.

The bar chart in Figure 2 shows the top 20 important features ranked by Random Forest, which reinforces the prominence of vibration and acoustic emission signals. Together, these findings validate that multi-domain vibration and AE-based features capture the most sensitive indicators of tool wear evolution in milling operations.

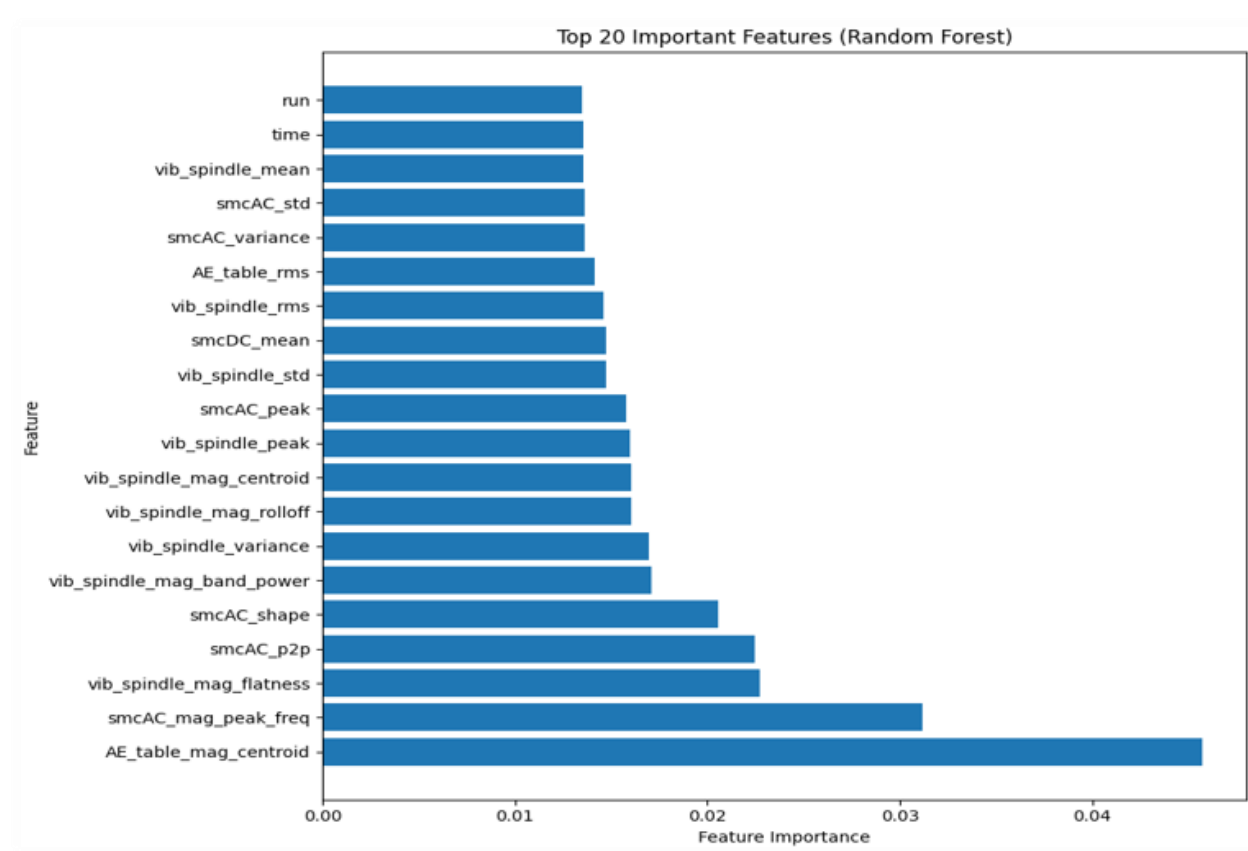


Figure 2: Top 20 important features derived from Random Forest classifier.

2.4 Predictive Modeling for Tool Wear Measure

Five supervised classification algorithms were developed and compared to predict tool wear levels from the ranked feature set. Each model was trained using 80% of the dataset and validated on the remaining 20%, employing accuracy, weighted precision, recall, and F1-score as key metrics.

Complexity Index (CI) and Effectiveness Index (EI) were also used to evaluate trade-offs between predictive accuracy and sensor usage efficiency.

2.4.1 Random Forest Classifier

The Random Forest algorithm is an ensemble-based method that constructs multiple decision trees during training and merges their outputs through majority voting [6]. Each tree is trained on a random subset of data and features, which reduces overfitting and improves generalization. This approach is particularly effective for high-dimensional datasets such as the NASA milling data, where non-linear relationships exist among sensor readings. In addition, Random Forest provides feature importance scores that help interpret which sensor variables contribute most to predicting tool wear.

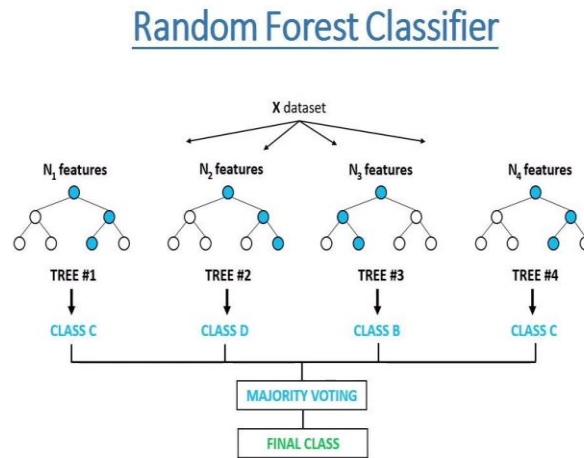


Figure 3: Random Forest classifier combining multiple decision trees through majority voting to predict the final class [7].

2.4.2 Support Vector Machine

The Support Vector Machine seeks to identify optimal hyperplanes that maximize the separation between different tool wear classes in a multidimensional space [8]. By applying kernel functions, the SVM can efficiently handle non-linear class boundaries that often occur in sensor-based manufacturing data. While computationally intensive for large datasets, SVMs are known for their strong classification performance and stability in smaller, well-structured datasets.

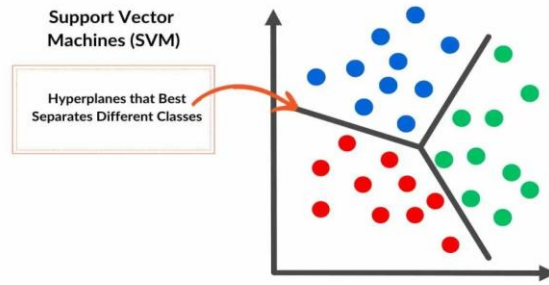


Figure 4: Support Vector Machine model showing hyperplanes that best separate different data classes [9].

2.4.3 Logistic Regression

Logistic Regression is a linear model that estimates the probability of class membership based on the logistic function [10]. It provides a straightforward and interpretable approach for mapping sensor features to wear classes. Although it assumes a linear relationship between predictors and the log-odds of the response, the model is computationally efficient and performs well when feature interactions are limited.

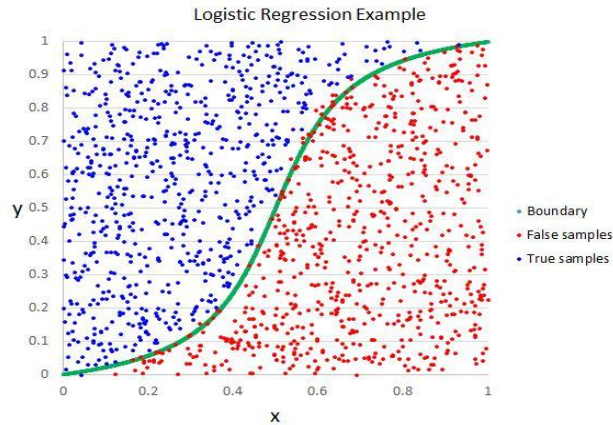


Figure 5: Logistic Regression model showing the boundary separating data classes [11].

2.4.4 Ensemble Model

The Ensemble model integrates the strengths of multiple base learners specifically Random Forest, Support Vector Machine, and Logistic Regression through a majority-voting mechanism. This hybrid approach capitalizes on the diversity of model architectures, combining tree-based, margin-based, and linear predictors to achieve improved generalization and stability. The Gaussian Naïve Bayes model was intentionally excluded from the ensemble due to its comparatively weak performance and poor handling of correlated sensor features.

2.4.5 Gaussian Naïve Bayes

The Gaussian Naïve Bayes algorithm applies Bayes' theorem under the assumption of conditional independence among input features, modeling each feature using a Gaussian (normal) distribution [12]. Its key advantage lies in computational simplicity, making it suitable for real-time or low-resource predictive applications. However, when sensor features are correlated as is common in tool wear data the independence assumption may reduce performance accuracy.

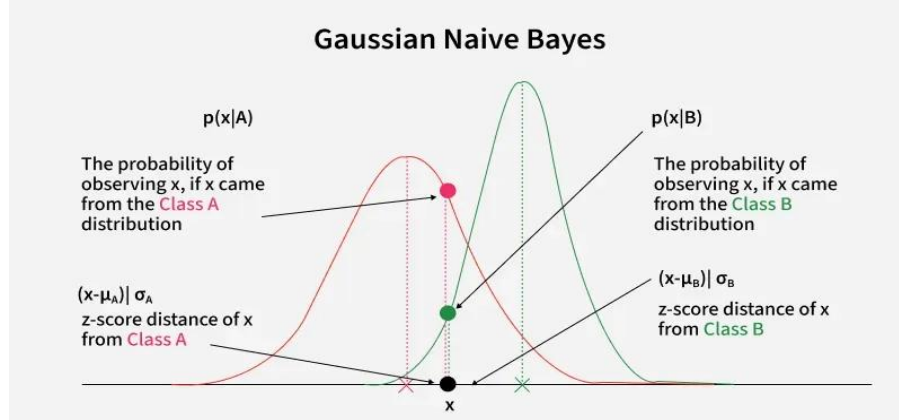


Figure 6: Illustration of Gaussian Naïve Bayes classification based on probability distributions of two classes [12].

2.5 Evaluation Metrics

Traditional evaluation metrics such as accuracy, precision, recall, and F1-score were used to assess the predictive performance of each model. In addition, customized indicators namely the Complexity Index (CI) and Effectiveness Index (EI) were introduced to evaluate model efficiency by balancing accuracy with sensor utilization [13]. The evaluation metrics can be defined as follows:

$$CI = \frac{\text{max sensors} - \text{sensors used}}{\text{max sensors} - \text{min sensors}}$$

$$EI = x \times \text{Accuracy} + (x - 1) \times CI$$

where **max sensors** is equal to the maximum number of sensors available in the dataset, **sensors used** are the number of sensors employed in a given model configuration, **min sensors** are the minimum number of sensors considered for evaluation, **Accuracy** is the overall prediction accuracy of the model, x is a weighting factor that balances the contribution of accuracy and complexity

These metrics were incorporated to capture the trade-off between predictive accuracy and system complexity, ensuring that high performance does not come at the cost of excessive sensor dependency or computational demand.

3. Result Analysis

This section presents the analysis of results obtained from the developed machine learning models. The performance of each predictive model is quantitatively evaluated using traditional and customized evaluation metrics, followed by an analysis of sensor-based ranking and the effect of increasing the number of classes on the model’s performance.

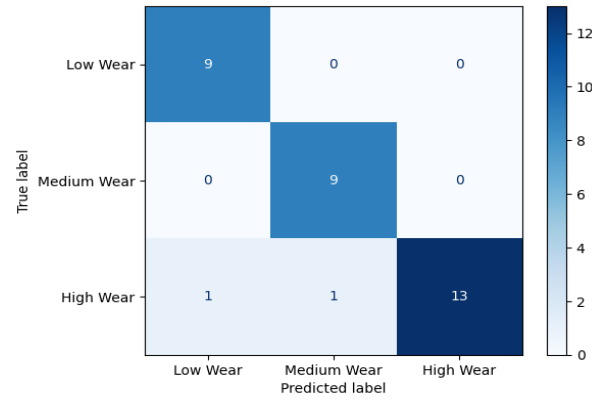
3.1 Quantitative Analysis of the Predictive Models

The quantitative performance of five predictive models was evaluated using the top 20 ranked features selected through the feature ranking method. A threshold value of 0.0134 in feature importance was applied to determine the most relevant variables irrespective of sensor origin, ensuring the best model performance. Since all models used data from the same four sensors, the Complexity Index (CI) remained constant at 0.4. The following table contains the quantitative performance of the models (the weight (x) was considered as 0.5 while calculating the Effectiveness Score):

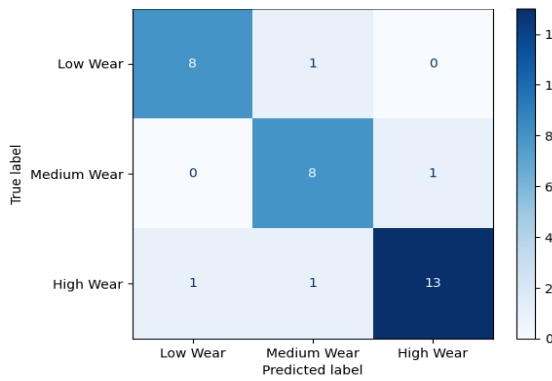
Table 3: Quantitative Performance of Predictive Models (bold font represents best performance)

	Accuracy	Precision	Recall	F1 Score	Effectiveness Index
Random Forest	93.94%	0.93	0.96	0.94	0.67
Support Vector Machine	87.88%	0.87	0.88	0.88	0.64
Logistic Regression	81.82%	0.87	0.79	0.79	0.61
Ensemble Model	87.88%	0.87	0.88	0.88	0.64
Gaussian Naïve Bayes	66.67%	0.68	0.65	0.64	0.54

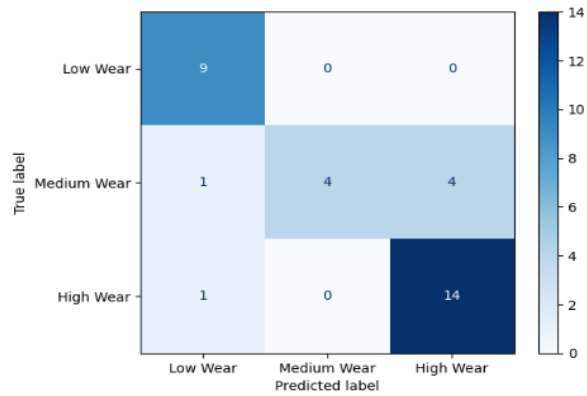
These metrics in the table reflect each model’s ability to accurately identify tool wear classes through precision, recall, and F1-score, while the Effectiveness Index (EI) accounts for both accuracy and complexity. Among them, Random Forest performed best, achieving high precision (0.93) and recall (0.96) due to its strength in capturing non-linear dependencies across features. The Ensemble and SVM models performed comparably, with the Ensemble model largely replicating the SVM’s decision-making pattern, showing similar accuracy and F1-scores.



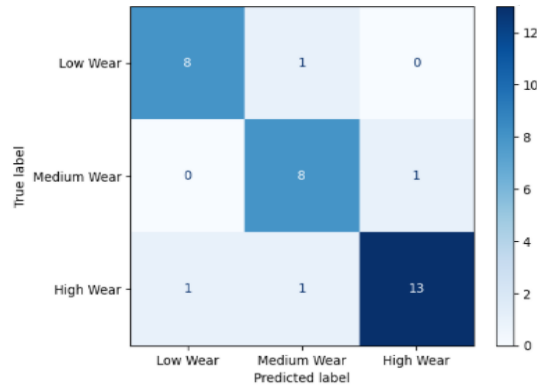
(a) Random Forest Classifier



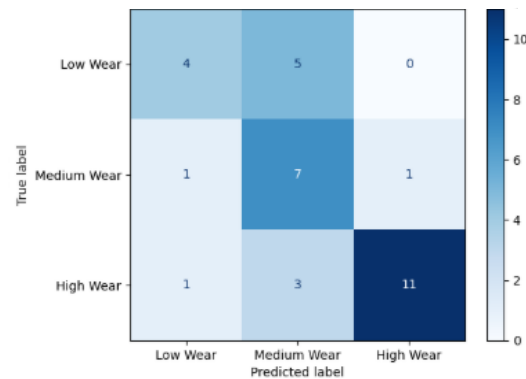
(b) Support Vector Machine



(c) Logistic Regression



(d) Ensemble Model



(e) Gaussian Naïve Bayes

Figure 7: Confusion matrices illustrating the class-wise performance of the five predictive models: (a) Random Forest Classifier, (b) Support Vector Machine, (c) Logistic Regression, (d) Ensemble Model and (e) Gaussian Naïve Bayes.

Logistic Regression, on the other hand, lagged due to its linear nature, making it less effective for capturing complex feature interactions present in the dataset. The Gaussian Naïve Bayes model showed the weakest results, as its assumption of feature independence does not hold true for

correlated sensor data typically found in machining processes. Overall, RF achieved the most balanced trade-off between accuracy and efficiency. The confusion matrices for all five models are presented below to illustrate class-wise prediction behavior.

Among all models in the confusion matrix, Random Forest achieved the best and most reliable predictions across all wear levels. The SVM and Ensemble models showed nearly identical classification patterns, with the Ensemble essentially mirroring SVM’s behavior. Logistic Regression performed moderately well, while Gaussian Naïve Bayes showed the weakest results due to its unrealistic assumption of feature independence. In terms of safe prediction, overestimating wear (e.g., predicting medium as high) is acceptable, but underestimating it (e.g., predicting medium as low) is more critical and can lead to tool failure, emphasizing the importance of safety-oriented evaluation beyond accuracy alone.

3.2 Ranking Sensors

To find a cost-effective solution that uses fewer sensors, we performed sensor ranking and selected sensors by their importance. Then, we build models incrementally.

Table 4: *Performance comparison of machine learning models using ranked sensor (Bold font indicates best performance)*

	1 sensor		2 sensors		3 sensors		4 sensors	
	Acc	EI	Acc	EI	Acc	EI	Acc	EI
Random Forest	96.97	0.97	93.94	0.94	87.80	0.60	93.94	0.40
Support Vector Machine	90.91	0.90	87.8	0.80	84.85	0.60	84.85	0.40
Logistic Regression	93.94	0.93	90.91	0.80	81.82	0.60	84.85	0.38
Ensemble Learning	93.94	0.94	90.91	0.80	87.88	0.60	87.88	0.36
Gaussian Naïve Bayes	57.58	0.57	51.52	0.51	51.52	0.51	54.55	0.35

First, we used the top-ranked sensor, then the next combination with other sensors. We also calculate the effectiveness index, which shows that models that using one sensor is more effective than others. The table below shows the performance of different machine learning models with ranked sensors. Here, we use ranked sensors using the feature importance score obtained from the feature ranking algorithm. Our results show that the model that uses the sensor “vib_spindle” performed better than combinations of other sensors in our experiments. Further, random forests performed better than other models in this case as well.

3.3 Multi-class Classification

We can build a simple solution using three classes. However, to capture the finer granularity of wear level, we divided VB values into six and eight classes. For class mapping, we took continuous but variable VB values and split them into 8 and 6 equal-sized groups (quantiles) using quantile binning. Each group contains roughly the same number of samples, from the smallest VB values to the largest. This converts continuous data into equal and ordered classes for accurate classification rather than manual division. We observe the distribution of classes using a box plot.

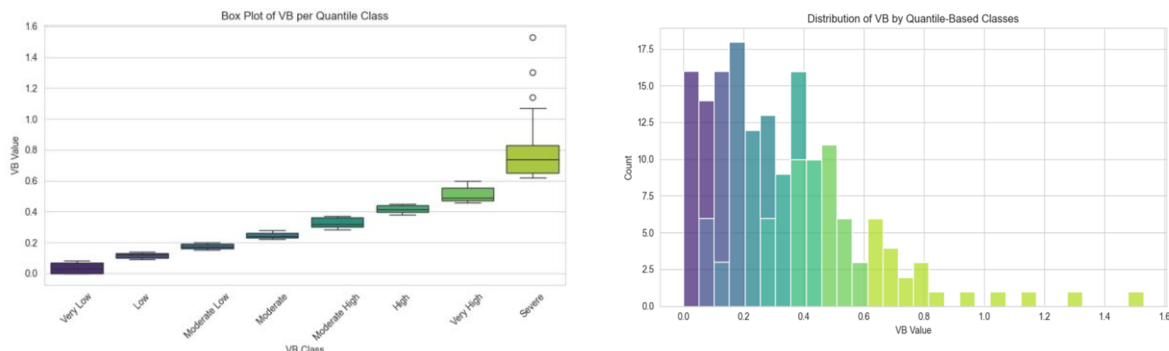


Figure 8: Distribution of VB values across multiple classes. The boxplot on the left states the spread and variability of VB values within each class, whereas the chart on the right displays quantile-based classification of VB values.

The plot shows the distribution of values within each class. Where each box represents the spread of VB values within that category. The box height shows how tightly or loosely the values are clustered. The short box represents the values that are close together (low variation), while the long box represents the values that are spread out (high variation). The clear upward trend shows that quantile mapping correctly orders the data. The upper end of our VB distribution is more widespread because extreme VB values pull the distribution upward. We also observe the distribution of VB by quantile-based classes. The graph shows it is right-skewed, that means VB has very few outliers.

Table 5: Performance comparison of machine learning models using N classes (Bold font represents best performance)

	N = 6			N = 8		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall
Random Forest	0.40	0.40	0.39	0.40	0.42	0.40
Support Vector Machine	0.36	0.32	0.36	0.28	0.24	0.28
Logistic Regression	0.38	0.39	0.38	0.30	0.35	0.30
Ensemble Learning	0.38	0.37	0.38	0.34	0.35	0.33

We observed that as the number of classes increases, the accuracy of the model drops. For three-class predictions, the results of models were more accurate than for 6 or 8 classes. In three-class predictions, the model just needs to learn a simple boundary. If we divide VB into 6 or 8 classes,

the model must learn more subtle distinctions between adjacent wear levels. The task becomes more difficult, so accuracy naturally decreases.

3.4 Safe vs Unsafe Classification

In our experiments, random forest works better than the other models. We observe that tool wear prediction is a nonlinear problem. Logistic regression assumes a linear decision boundary, so it struggles with nonlinear patterns. SVM with a non-linear kernel also fails nonlinear separations. However, random forest captures nonlinear interactions because it splits the feature space recursively in decision trees.

3.4.1 Confusion Matrix

The confusion matrix shows that the model achieves good separation between classes (Very Low – Moderate Low) and (High – Very High), but the middle transition zones are less distinct. We observe that the area above the diagonal in the confusion matrix is a safe zone, while the lower area represents an unsafe zone. If we look at our best model, random forest, we can observe that most of the predictions are in the safe zone. However, we think there is a need for improvements to make predictions safer.

1. Random Forest:

The confusion matrix of random forest in **Figure 9** shows that the ‘very low’, ‘very high’, and ‘severe’ are the best performing classes among the other classes.

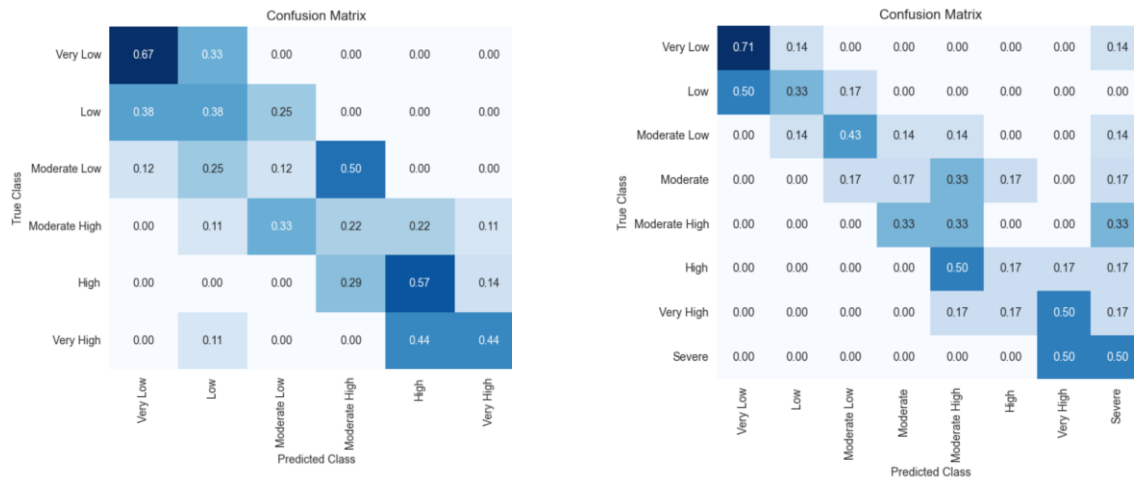


Figure 9: Confusion Matrix for Random Forest on 6 classes (Left) and 8 Classes (Right)

2. Support Vector Machine:

For the six-class confusion matrix, very low and very high, while for the eight-class confusion matrix best classes are very low, moderate low were the best performing classes. This shows that the model needs to be better to predict the severe class because the severe class is important to predict tool wear.

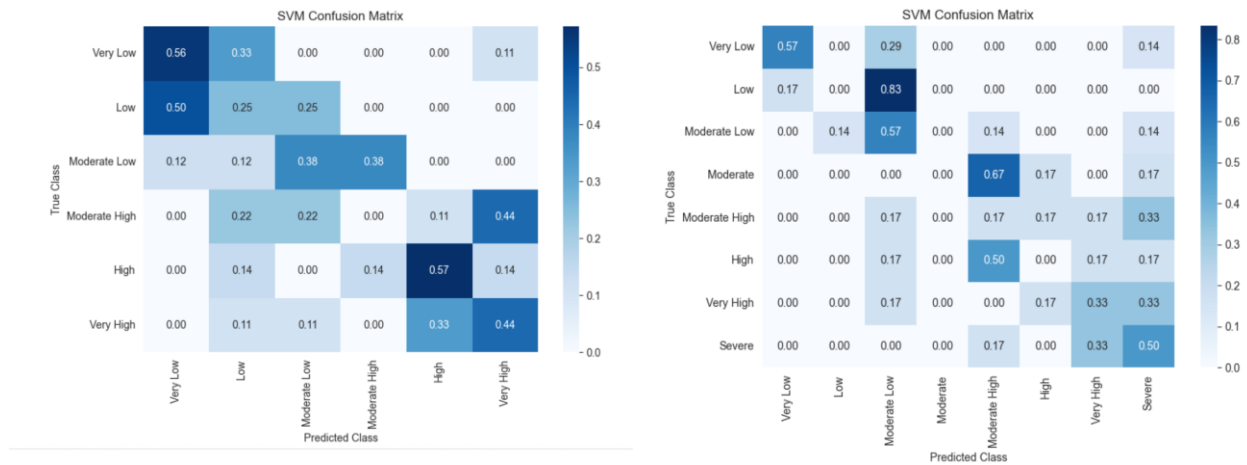


Figure 10: Confusion Matrix for Support Vector Machine on 6 classes (Left) and 8 Classes (Right)

3. Logistic Regression:

Figure 11 shows that the six-class confusion matrix model performs well for the high class, while the eight-class confusion matrix has severe class as the best performing class. The confusion matrix shows that the model needs improvement in other classes to be considered as best predictive model.

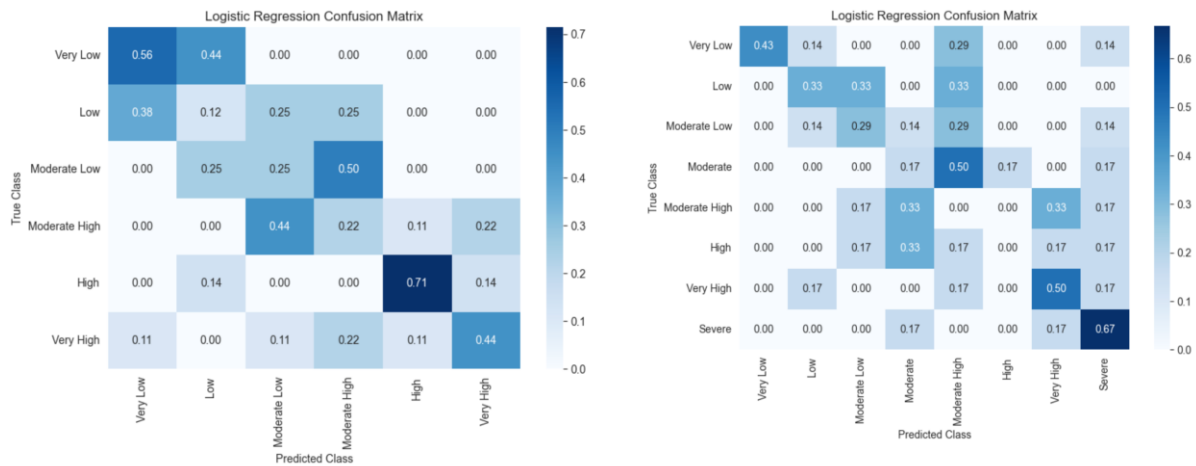


Figure 11: Confusion Matrix for Logistic Regression on 6 classes (Left) and 8 Classes (Right)

4. Ensemble Learning:

The confusion matrix in Figure 12 shows that ensemble models are considerably better than SVM and logistic regression, as it has various best-performing classes. The severe class in both six and eight-class confusion matrices is also the best working classes, which is crucial for correct tool wear prediction. Severe class is important because it shows that the tool is very close to wear.

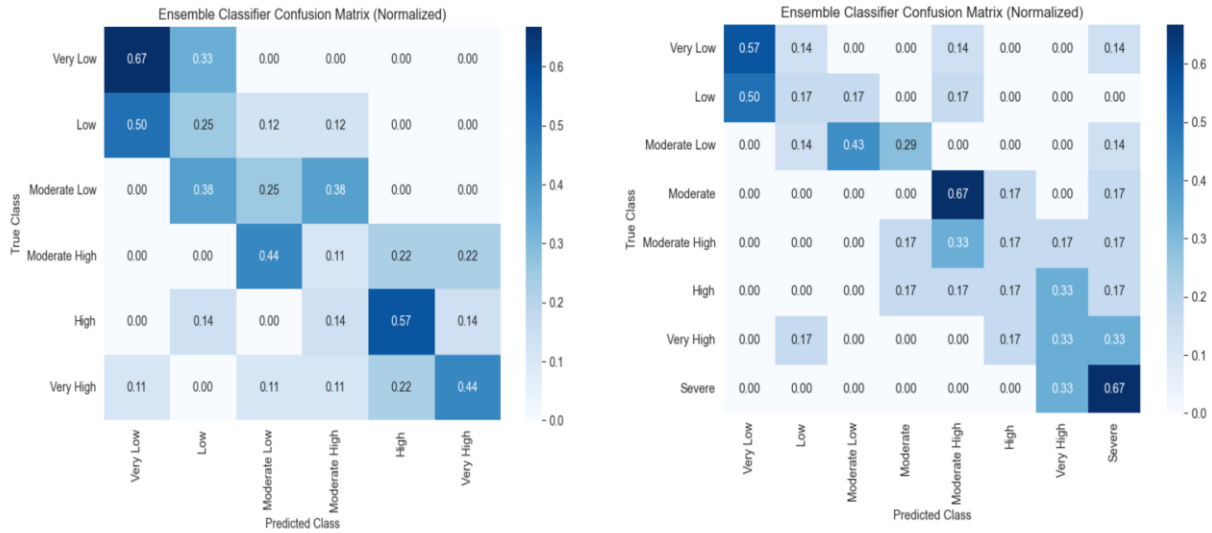


Figure 12: Confusion Matrix for Ensemble Model on 6 classes (Left) and 8 Classes (Right)

4. Conclusion

In conclusion, this study focuses on tool wear prediction in machining operations, examining the effects of feature ranking, sensor ranking, and increasing class granularity on model performance. The results showed that good feature ranking significantly improved the model's predictive accuracy, enabling them to identify key variables that contribute most to tool wear. Among the tested models, the Random Forest Classifier achieved the highest accuracy and stability, followed by the SVM and Ensemble models, while Logistic Regression performed moderately, and Gaussian Naïve Bayes showed the weakest results due to its oversimplified assumptions. Sensor ranking further demonstrated that even with a minimal number of sensors, the models could still maintain high accuracy, leading to a strong overall effectiveness score. However, as the number of wear classes increased, the model's accuracy declined, indicating a trade-off between prediction accuracy and classification granularity. Overall, Random Forest provides the most effective, efficient, and practical framework for real-time, data-driven tool wear prediction.

Contribution

Ahmed conducted literature reviews, motivation, evaluation metrics design, and dataset analysis. Zinah handled data preprocessing, feature ranking, sensor ranking, and feature analysis. Prमित handled the predictive model design for tool wear and evaluation for 5 models. Suleman divided VB values into ($n = 6$ and 8 classes), investigated the impact of changing the number of classes, and built models with sensor ranking to find a cost-effective solution.

References:

- [1] Yao K-C, Chen D-C, Pan C-H, Lin C-L. The Development Trends of Computer Numerical Control (CNC) Machine Tool Technology. *Mathematics*. 2024; 12(13):1923. <https://doi.org/10.3390/math12131923>
- [2] Hirsch, E., & Friedrich, C. Data-driven tool wear prediction in milling, based on a process-integrated single-sensor approach. 2025; arXiv preprint arXiv:2412.19950. <https://doi.org/10.48550/arXiv.2412.19950>
- [3] Hegab H, “Predicting Tool Wear in Manufacturing Operations Using Machine Learning,” *UNIV*6090 – CourseLink*. 2025. [Online]. Available: <https://courselink.uoguelph.ca/d2l/le/content/980581/viewContent/4271703/View>. [Accessed: November 4, 2025]
- [4] K. Goebel and A. Agogino, “Milling Data Set,” NASA Ames Prognostics Data Repository, Moffett Field, CA, 2007.
- [5] GeeksforGeeks, “ZScore Normalization: Definition and Examples,” GeeksforGeeks, Jul. 26, 2024. <https://www.geeksforgeeks.org/data-analysis/z-score-normalization-definition-and-examples/>
- [6] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [7] A. Chauhan, “Random Forest Classifier and its Hyperparameters,” *Analytics Vidhya – Medium*, Feb. 23, 2021. [Online]. Available: <https://medium.com/analytics-vidhya/random-forest-classifier-and-its-hyperparameters-8467bec755f6>. [Accessed: November 6, 2025]
- [8] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995, doi: 10.1007/BF00994018.
- [9] N. V. Otten, “Support Vector Machines (SVM) In Machine Learning Made Simple & How To Tutorial,” *Spot Intelligence*, 6 May 2024. [Online]. Available: <https://spotintelligence.com/2024/05/06/support-vector-machines-svm/>. [Accessed: 6 Nov. 2025].
- [10] J. S. Cramer, “The origins of logistic regression,” *Tinbergen Institute Working Paper*, no. 2002-119/4, Dec. 2002, doi: 10.2139/ssrn.360300.
- [11] A. Vianna, “How to Implement Logistic Regression Algorithm, with Orange Data Science Tool,” *My Data Science Journey – Medium*, 6 Nov. 2021. [Online]. Available: <https://medium.com/my-data-science-journey/how-to-implemente-logistic-regression-algorithm-in-machine-learning-dd9834e1c89b>. [Accessed: 6 Nov. 2025].
- [12] S. Joshdylh, “Gaussian Naive Bayes,” GeeksforGeeks, 28 Aug. 2025. [Online]. Available: <https://www.geeksforgeeks.org/machine-learning/gaussian-naive-bayes/>. [Accessed: 02-Nov-2025].
- [13] M. Abubakr, M. A. Hassan, G. M. Krolczyk, N. Khanna, and H. Hegab, “Sensors selection for tool failure detection during machining processes: A simple accurate classification model,” *CIRP Journal of Manufacturing Science and Technology*, vol. 32, pp. 108–119, 2021, doi: 10.1016/j.cirpj.2020.12.002.